MOHAMMAD AZAM

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Portfolio: mohammadazam7.github.io/mohammadazam7

Summary

Data & Software Engineer with 4 years of experience in data engineering, analytics, ML, and production software. Builds ETL pipelines, dashboards, ML models, retrieval systems, and tested APIs with CI/CD in Python, SQL, and C#/.NET on AWS and Azure.

Technical Skills

Languages:	Python, SQL, C++, C#/.NET, Java, JavaScript, TypeScript, Django, Spring Boot, React
Data Engineering:	ETL/ELT, Spark (PySpark), dbt, Airflow, Kafka, data modeling (star schema)
Data Platforms:	Snowflake, Databricks, Redshift, PostgreSQL, SQL Server, MongoDB
Analytics & BI:	Power BI, Tableau, Pandas, statistical analysis, KPI dashboards
ML & AI:	scikit-learn, TensorFlow, PyTorch, feature engineering, LLM APIs, RAG
Cloud & DevOps:	AWS (S3, Glue, Lambda), Azure (Data Factory), Docker, Kubernetes, Git, CI/CD
Soft Skills:	Communication, Collaboration, Ownership, Problem-Solving, Curiosity

Experience

Data Engineer, Analytics & BI | FGF Brands, Toronto, ON, Canada — Jan 2026 – Present
- Built Python and SQL pipelines loading 5 source systems (50K rows/day) into reporting tables, saving 10 hours/week
- Added validation checks that caught over 100 data errors before they reached 4 teams
- Delivered Power BI reports for multiple plants, cutting report turnaround from 2 days to 2 hours
- Tuned slow SQL with execution plans, CTEs, and window functions to speed up daily reporting

Data & Software Engineer | Konarak Global Services, Rudrapur, Uttarakhand, India — Nov 2021 – Nov 2024
- Developed Python, C#/.NET, and Java services and REST APIs for 3 production client applications
- Engineered features for TensorFlow and scikit-learn models, raising prediction accuracy from 82% to 90%
- Set up Azure DevOps CI/CD with 150 automated tests per pull request, cutting release time from 1 day to under 1 hour
- Traced a production memory leak to its root cause, ending weekly crashes and cutting latency 30%

AI/ML Engineer Intern | Midas Consulting, Noida, UP, India — Jul 2021 – Oct 2021
- Wrote SQL and Python ETL scripts to clean and prepare EHR data, engineering features such as patient census and shift history
- Improved staffing accuracy by 25% with predictive models in Python, TensorFlow, and SQL
- Cut overtime costs by 15% via nurse-shift optimization; presented forecasts to hospital operations teams
- Launched real-time dashboards to track staffing levels and overtime trends for operations staff

Education

Master of Applied Computing (AI Stream) | University of Windsor, Windsor, ON, Canada | 80% average — Jan 2025 – Aug 2026
B.Tech in Computer Science Engineering | Glocal University, Saharanpur, UP, India — Jun 2017 – May 2021

Projects

RAG Evaluation Platform | GitHub | *Python, BM25, Embeddings, FastAPI, NumPy, pytest* — Oct 2026 – Present
- Building cited question answering over a document collection, with an evaluation harness that judges every pipeline change
- Split documents into passages with stable IDs and exact offsets, so each answer traces back to its source text
- Combined BM25 and dense retrieval with reciprocal rank fusion; next, scoring recall@k, MRR, and faithfulness

NYC Taxi Data Lake on AWS | GitHub | *Python, PySpark, AWS Glue, S3, Athena, Redshift* — Sep 2026 – Oct 2026
- Created an AWS batch pipeline ingesting NYC taxi trip data (about 3M rows/month) into an S3 data lake, cataloged with Glue
- Authored PySpark Glue jobs to clean, deduplicate, and validate trips, loading a star schema into Redshift Serverless
- Configured data-quality checks with CloudWatch and SNS alerts, plus a dashboard of trip volume, revenue, and busiest zones

AI Multimodal Assistant | GitHub | *Python, FastAPI, React, PyTorch, OpenCV, Docker, Kubernetes* — Aug 2025 – Sep 2025
- Built a FastAPI service running YOLOv8 detection on live camera streams, with a React client drawing results
- Added cross-stream micro-batching and latest-frame back-pressure, holding p95 latency under 1 s at 4x overload
- Deployed on Kubernetes with Docker and autoscaling on CPU and open streams; benchmarked with a load-test harness

Credify | GitHub | *Java, Spring Boot, React, JavaScript, MySQL, Selenium* — Mar 2025 – Apr 2025
- Architected a full-stack app (Spring Boot, React, MySQL) comparing credit cards across Canadian and U.S. banks
- Implemented multi-criteria search and filtering with query response times under 200 ms
- Introduced Selenium UI tests to catch front-end bugs before release

Extracurricular Activities

First Place, COMP 8967 Pitch Competition | University of Windsor — Fall 2025
Demo Day Presenter, 2nd Place | University of Windsor — Apr 2025
- Presented an AI-driven project, awarded 2nd place by industry judges for innovation and technical excellence

Vice President, Computer Science Society | Glocal University | Led campus tech events — Aug 2019 – Sep 2020